



PROBLEM SESSION Week #3: Process Manipulation and Monitoring.

1) Learn the important Linux commands for manipulating and monitoring the process or jobs

- 1) Discuss the difference of “top” and “ps” command.
- 2) What is the purpose of the “nice” command? How can you nice a process?
- 3) How to kill the processes and/or jobs? Give example in the report.

2) Discuss these situations

First situation

- 1) Create a shell script (just a text file), named ss1_1.sh.
- 2) In the script, the “sleep” command is called with 10 second.
- 3) Run the script ss1_1.sh (./ss1_1.sh), wait for 10 second
- 4) Run the script ss1_1.sh again with the sigh “&” in the end of command (./ss1_1.sh &)
- 5) Discuss the difference of both running methods. (3 and 4)

NOTE: If your text file is not runnable, use ‘chmod’ command to fix the issue.

You could try to use a for loop in script and sleep one second per iteration.

Second situation

- 1) Create a shell script, named ss1_2.sh.
- 2) In the script, the “sleep” command is called with 1000 second.
- 3) Run the script ss1_2.sh (./ss1_2.sh), then press CTRL+Z.
- 4) Use “jobs” command, see the result and discuss the result of pressing CTRL+Z
- 5) Show how to use “fg” and “bg” commands to resume the script ss1_2.sh and discuss the difference of both resume commands.

3) Find and try out process monitoring tools with GUI on Linux system

- 1) What is it?
- 2) How to acquire it?
- 3) How to use it?
- 4) What value it can monitor?

4) Find and try out process monitoring tools on Windows system

- 1) What is it?
- 2) How to acquire it?
- 3) How to use it?
- 4) What value it can monitor?

- 5) Write C function to simulate the **STCF**, the program 'PS3.c' that is already provided will read the data in .csv format and convert it into a struct, before sending it to your function, which must process this struct and show which process is running for each time slot.

Assume:

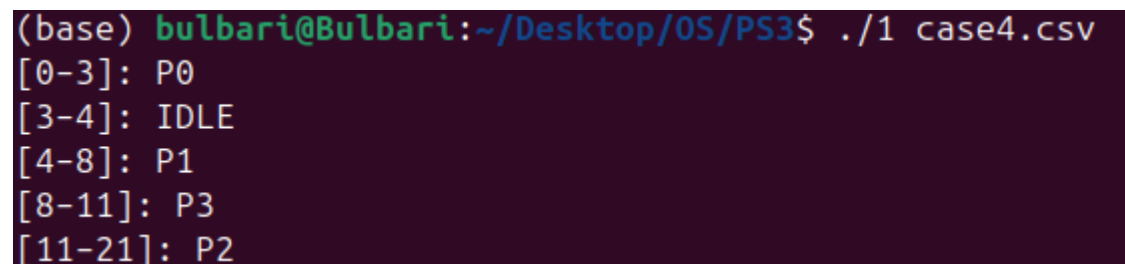
- Your function and context switch occur very fast. You shouldn't count this as CPU usage.
- The process that arrives at time T means "arrives at the very end of time T-1, but before the scheduler updates the ready queue for time T"
- If multiple processes arrive at the same time, sort by using process_id.'
- If multiple processes have the same remaining time, sort by using process_id.'

Note that in the original code, the file 'case1.csv', case2.csv', and 'case3.csv' must be in the same directory as the executable file. And it can be used via the terminal, for example

```
./PS3 case1.csv
```

All of the input cases and the code that processes them are already in LEB2. You may use the PS3.c and fill in the provided blank space in the function with your version of the scheduler.

The expected result that your program needed to print out should look like this:



```
(base) bulbari@Bulbari:~/Desktop/OS/PS3$ ./1 case4.csv
[0-3]: P0
[3-4]: IDLE
[4-8]: P1
[8-11]: P3
[11-21]: P2
```

Show the screenshot of your function, with the overview explanation + screenshot of 3 cases results + discussion.

Submission

Submit the document file into LEB2 before due date. This is the problem session for the group of up to 3-4 students. No need to submit any code file.